State Space Representations of Linear Physical Systems

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Introduction

As systems become more complex, representing them with differential equations or transfer functions becomes cumbersome. This is even more true if the system has multiple inputs and outputs. This document introduces the state space method which largely alleviates this problem. The state space representation of a system replaces an nth order differential equation with a single first order *matrix* differential equation. The state space representation of a system is given by two equations :

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t)$$
$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{u}(t)$$

*Note: Bold face characters denote a vector or matrix.The variable **x** is more commonly used in textbooks and other references than is the variable **q** when state variables are discussed. The variable **q** will be used here since we will often use **x** to represent position.*

The first equation is called the state equation, the second equation is called the output equation. For an nth order system (i.e., it can be represented by an nth order differential equation) with *r* inputs and *m* outputs the size of each of the matrices is as follows:

- **q** is $n \times 1$ (n rows by 1 column); **q** is called the state vector, it is a function of time
- **A** is $n \times n$; **A** is the state matrix, a constant
- **B** is $n \times r$; **B** is the input matrix, a constant
- **u** is $r \times 1$; **u** is the input, a function of time
- **C** is $m \times n$; **C** is the output matrix, a constant
- **D** is $m \times r$; **D** is the direct transition (or feedthrough) matrix, a constant
- **y** is $m \times 1$; **y** is the output, a function of time

Note several features:

- The state equation has a single first order derivative of the state vector on the left, and the state vector, **q**(t), and the input **u**(t) on the right. There are no derivatives on the right hand side.
- The output equation has the output on the left, and the state vector, **q**(t), and the input **u**(t) on the right. There are no derivatives on the right hand side.

For systems with a single input and single output (i.e., most of the systems we will consider) these variables become (with r=1 and m=1):

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}u(t)$$
$$y(t) = \mathbf{C}\mathbf{q}(t) + Du(t)$$

where

- **q** is n*x*1 (n rows by 1 column)
- **A** is n*x*n
- **B** is n*x*1
- u is 1*x*1 (i.e., a scalar)
- **C** is 1*x*n
- D is 1*x*1 (i.e, a scalar)
- y is 1*x*1 (i.e, a scalar)

Advantages of this representation include:

- The notation is very compact. Even large systems can be represented by two simple equations.
- Because all systems are represented by the same notation, it is very easy to develop general techniques to solve these systems.
- Computers easily simulate first order equations.

A Simple Example

Consider an 4th order system represented by a single 4th order differential equation with input x and output y.

$$\dddot{y} + a_1\ddot{y} + a_2\dot{y} + a_3\dot{y} + a_4 y = b_0 x$$

We can define 4 new variables, q1 through q4.

$$q_1 = y$$
$$q_2 = \dot{q}_1 = \dot{y}$$
$$q_3 = \dot{q}_2 = \ddot{y}$$
$$q_4 = \dot{q}_3 = \dddot{y}, \quad \text{so}$$
$$\dddot{y} + a_1 q_4 + a_2 q_3 + a_3 q_1 + a_4 q_1 = b_0 x$$

but

$$\dot{q}_4 = \dddot{y}, \quad \text{so}$$
$$\dddot{y} = \dot{q}_4 = -a_4 q_1 - a_3 q_2 - a_2 q_3 - a_1 q_4 + b_0 x$$

We can now rewrite the 4th order differential equation as 4 first order equations

$$\dot{q}_1 = q_2 = \dot{y}$$
$$\dot{q}_2 = q_3 = \ddot{y}$$
$$\dot{q}_3 = q_4 = \dddot{y}$$
$$\dot{q}_4 = -a_4 q_1 - a_3 q_2 - a_2 q_3 - a_1 q_4 + b_0 x$$

This is compactly written in state space format as

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u \qquad \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -a_4 & -a_3 & -a_2 & -a_1 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ b_0 \end{bmatrix} u$$

$$y = \mathbf{C}\mathbf{q} + Du \qquad y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$$

with

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -a_4 & -a_3 & -a_2 & -a_1 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ b_0 \end{bmatrix}; \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}; \quad D = 0; \quad \text{and the input} = u = x$$

For this problem a state space representation was easy to find. In many cases (e.g., if there are derivatives on the right side of the differential equation) this problem can be much more difficult. Such cases are explained in the discussion of transformations between system representations.

The state space representation is not unique

Case 1: Alternate State Space Representation

Another important point is that the state space representation is not unique. As a simple example we could simply reorder the variables from the example above (the new state variables are labeled $\mathbf{q}_{new}$). This results in a new state space representation

$$q_{3,new} = y$$
$$\dot{q}_{3,new} = q_{2,new} = \dot{y}$$
$$\dot{q}_{2,new} = q_{1,new} = \ddot{y}$$
$$\dot{q}_{1,new} = q_{4,new} = \dddot{y}$$

$$\dot{q}_{4,new} = -a_4 q_{3,new} - a_3 q_{2,new} - a_2 q_{1,new} - a_1 q_{4,new} + b_0 x$$

$$\dot{\mathbf{q}}_{new} = \mathbf{A}_{new}\mathbf{q}_{new} + \mathbf{B}_{new}u \qquad \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -a_2 & -a_3 & -a_4 & -a_1 \end{bmatrix} \begin{bmatrix} q_{1,new} \\ q_{2,new} \\ q_{3,new} \\ q_{4,new} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ b_0 \end{bmatrix} u$$

$$y = \mathbf{C}_{new}\mathbf{q}_{new} + D_{new}u \qquad y = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} q_{1,new} \\ q_{2,new} \\ q_{3,new} \\ q_{4,new} \end{bmatrix}$$

$$\mathbf{A}_{new} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -a_2 & -a_3 & -a_4 & -a_1 \end{bmatrix}; \quad \mathbf{B}_{new} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ b_0 \end{bmatrix};$$
$$\mathbf{C}_{new} = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}; \quad D_{new} = 0; \quad \text{and the input} = u = x$$

Case 2: Alternate State Space Representations

In the previous case careful examination of the original and modified state space system reveals that they represent the same system. However we can make entirely new state variables by forming linear combination of the original state variables in which this equality is not obvious. Consider the state variable $\mathbf{q}_{new}$ defined as follows:

$$q_{1,new} = y - \dot{y}$$
$$q_{2,new} = y + \dot{y}$$
$$q_{3,new} = \ddot{y}$$
$$q_{4,new} = \dddot{y}$$

In this case the new state space variables are given by (*the details of how these matrices are determined are not important for this discussion. They are given here if you are interested*):

$$\dot{\mathbf{q}}_{new} = \mathbf{A}_{new}\mathbf{q}_{new} + \mathbf{B}_{new}u$$
$$y = \mathbf{C}_{new}\mathbf{q}_{new} + D_{new}u$$

$$\mathbf{A}_{new} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & -1 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{a_3 - a_4}{2} & -\frac{a_3 + a_4}{2} & -a_2 & -a_1 \end{bmatrix}$$

$$\mathbf{B}_{new} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ b_0 \end{bmatrix}$$

$$\mathbf{C}_{new} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$$D_{new} = 0$$

This new state space system is quite different from the original one, and it is not at all obvious that they represent the same system. (*It can be shown that the systems are identical by transforming the state space representation to a transfer function. Techniques for doing so are discussed elsewhere.*)

Key Concept: Defining a State Space Representation

A nth order linear physical system can be represented using a state space approach as a single first order matrix differential equation:

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t)$$
$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{u}(t)$$

The first equation is called the state equation and it has a first order derivative of the state variable(s) on the left, and the state variable(s) and input(s), multiplied by matrices, on the right. The second equation is called the output equation and it has teh output on the left and the state variable(s) and input(s), multiplied by matrices, on the right. No other terms are allowed in the equation. In these equations:

- **q** is n*x*1 (n rows by 1 column); **q** is called the state vector, it is a function of time
- **A** is n*x*n; **A** is the state matrix, a constant
- **B** is n*x*r; **B** is the input matrix, a constant
- **u** is r*x*1; **u** is the input, a function of time
- **C** is m*x*n; **C** is the output matrix, a constant
- **D** is m*x*r; **D** is the direct transition (or feedthrough) matrix, a constant
- **y** is m*x*1; **y** is the output, a function of time

For a single input, single output system (the case that interests us the most):

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}u(t)$$
$$y(t) = \mathbf{C}\mathbf{q}(t) + Du(t)$$

- **q** is n*x*1 (n rows by 1 column)
- **A** is n*x*n
- **B** is n*x*1
- u is 1*x*1 (i.e., a scalar)
- **C** is 1*x*n
- D is 1*x*1 (i.e, a scalar)
- y is 1*x*1 (i.e, a scalar)

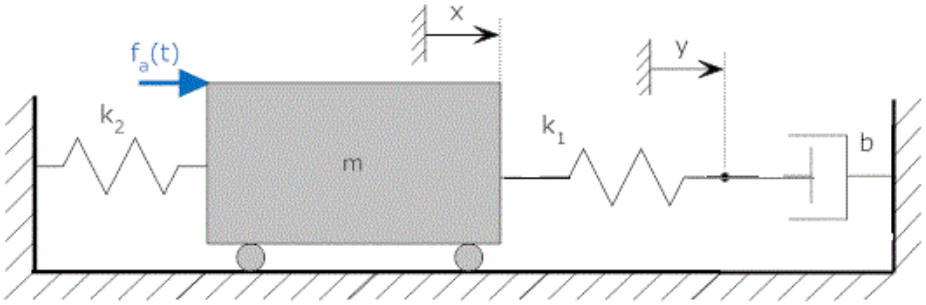
The state space representation is not unique; many (actually an infinite number) of state space systems can be used to represent any linear physical system.

Developing a state space model from a system diagram (Mechanical Translating)

Another, powerful, way to develop a state space model is directly from the free body diagrams. If you choose as your state variables those quantities that determine the energy in the system, a state space system is often easy to derive. For example, in a mechanical system you would choose extension of springs (potential energy, ½kx²) and the velocity of masses (kinetic energy, ½mv²); for electrical systems choose voltage across capacitors, ½Ce² (e=voltage)) and current through inductors (½Li²). This is best illustrated by several examples, two rotating and one electrical.

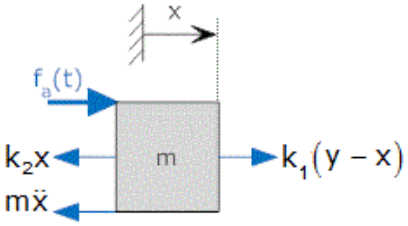
Example: Direct Derivation of State Space Model (Mechanical Translating)

Derive a state space model for the system shown. The input is f_a and the output is y.

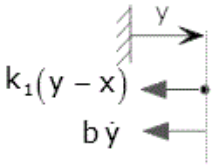


We can write free body equations for the system at x and at y.

Freebody Diagram **Equation**



$$m\cdot\ddot{x}+k_1\cdot x+k_2x-k_1\cdot y=f_a$$



$$b\cdot\dot{y}+k_1y-k_1\cdot x=0$$

There are three energy storage elements, so we expect three state equations. The energy storage elements are the spring, k_2, the mass, m, and the spring, k_1. Therefore we choose as our state variables x (the energy in spring k_2 is $½k_2x^2$), the velocity at x (the energy in the mass m is ½mv², where v is the first derivative of x), and y (the energy in spring k_1 is $½k_1(y-x)^2$, so we could pick y-x as a state variable, but we'll just use y (since x is already a state variable; recall that the choice of state variables is not unique). Our state variablesbecome:

$$q_1=x$$
$$q_2=\dot{x}$$
$$q_3=y$$

Now we want equations for their derivatives. The equations of motion from the free body diagrams yield

$$\dot{q}_1=\dot{x}=q_2$$
$$\dot{q}_2=\ddot{x}=\frac{1}{m}\left(f_a-k_1x-k_2x+k_1y\right)$$
$$=\frac{1}{m}\left(f_a-k_1q_2-k_2q_1+k_1q_3\right)$$
$$\dot{q}_3=\dot{y}=\frac{k_1}{b}(x-y)=\frac{k_1}{b}(q_1-q_3)$$

or

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u \qquad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{k_1 + k_2}{m} & 0 & \frac{k_1}{m} \\ \frac{k_1}{b} & 0 & -\frac{k_1}{b} \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \end{bmatrix}$$

$$y = \mathbf{C}\mathbf{q} + Du \qquad \mathbf{C} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \qquad D = 0$$

with the input $u=f_a$.

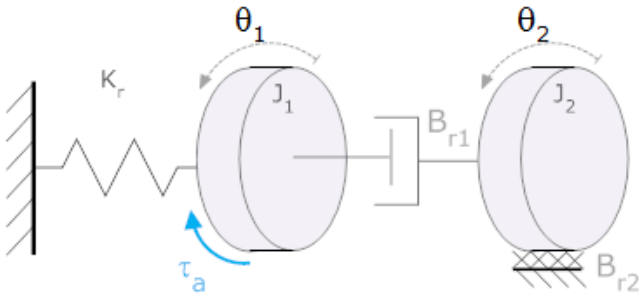
This technique does not always easily yield a set of state equations (read about some examples here). In some cases it is easier to develop a transfer function model and convert this to a state space model. Transfer functions are discussed elsewhere.

Developing state space model from system diagram (Mechanical Rotating)

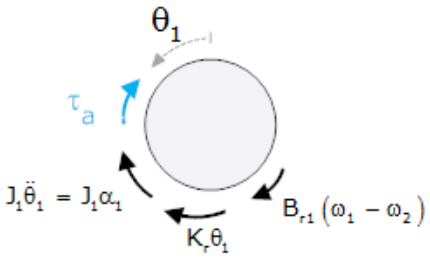
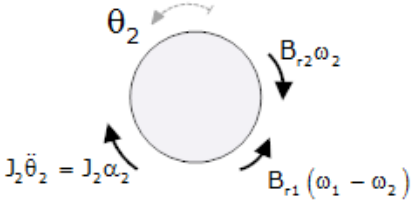
The energy variables for rotating systems are potential energy stored in springs ($\frac{1}{2}K_r\theta^2$) and kinetic energy stored in inertial elements ($\frac{1}{2}J\omega^2$).

Example: Direct Derivation of State Space Model (Mechanical Rotating)

Derive a state space model for the system shown. The input is τ_a and the output is θ_1.



We can write free body equations for the system at θ_1 and θ_2

Freebody Diagram	Equation
	$J_1\ddot{\theta}_1 + B_{r1}\dot{\theta}_1 + K_r\theta_1 - B_{r1}\dot{\theta}_2 = -\tau_a$
	$J_2\ddot{\theta}_2 + (B_{r2} + B_{r1})\dot{\theta}_2 - B_{r1}\dot{\theta}_1 = 0$

There are three energy storage elements, so we expect three state equations. Energy is stored as potential energy in the spring ($\frac{1}{2}K_r\theta_1^2$) and kinetic energy in the two flywheels ($\frac{1}{2}J_1\alpha_1^2$, $\frac{1}{2}J_2\alpha_2^2$). Our state variable equations become:

$$q_1 = \theta_1$$
$$q_2 = \dot{\theta}_1$$
$$q_3 = \dot{\theta}_2$$

Now we want equations for their derivatives. The equations of motion from the free body diagrams yield

$$\dot{q}_1 = \dot{\theta}_1 = q_2$$
$$\dot{q}_2 = \ddot{\theta}_1 = \frac{1}{J_1}\left(-\tau_a - B_{r1}\dot{\theta}_1 - K_r\theta_1 + B_{r1}\dot{\theta}_2\right)$$
$$= -\frac{1}{J_1}\tau_a - \frac{B_{r1}}{J_1}q_2 - \frac{K_r}{J_1}q_1 + \frac{B_{r1}}{J_1}q_3$$
$$\dot{q}_3 = \ddot{\theta}_2 = \frac{1}{J_2}\left(-(B_{r2}+B_{r1})\dot{\theta}_2 + B_{r1}\dot{\theta}_1\right)$$
$$= -\frac{(B_{r2}+B_{r1})}{J_2}q_3 + \frac{B_{r1}}{J_2}q_2$$

or

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u \quad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{K_r}{J_1} & -\frac{B_{r1}}{J_1} & \frac{B_{r1}}{J_1} \\ 0 & \frac{B_{r1}}{J_2}q_2 & -\frac{(B_{r2}+B_{r1})}{J_2} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ -\frac{1}{J_1} \\ 0 \end{bmatrix}$$
$$y = \mathbf{C}\mathbf{q} + Du \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \quad \mathbf{D} = 0$$

with the input $u=\tau_a$, and the output $y=\theta_1$.

Developing State space model from system diagram (Electrical)

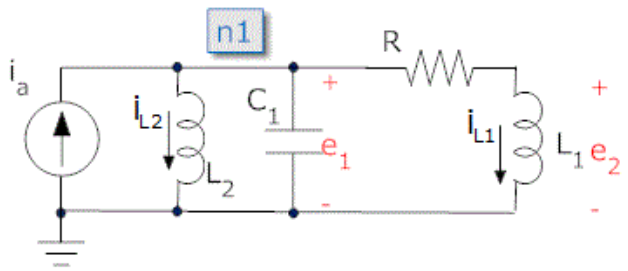
To develop a state space system for an electrical system, they choosing the voltage across capacitors, and current through inductors as state variables. Recall that

$$e_{inductor} = L\frac{di_{inductor}}{dt}$$
$$i_{capacitor} = C\frac{de_{capacitor}}{dt}$$

so if we can write equations for the voltage across an inductor, it becomes a state equation when we divide by the inductance (i.e., if we have an equation for $e_{inductor}$ and divide by L, it becomes an equation for $di_{inductor}/dt$ which is one of our state variable). Likewise if we can write an equation for the current through the capacitor and divide by the capacitance it becomes a state equation for $e_{capacitor}$. This is best illustrated by an example.

Example: Direct Derivation of State Space Model (Electrical)

Derive a state space model for the system shown. The input is i_a and the output is e_2.



There are three energy storage elements, so we expect three state equations. Try choosing i_1, i_2 and e_1 as state variables. Now we want equations for their derivatives. The voltage across the inductor L_2 is e_1 (which is one of our state variables)

$$L_2 \frac{di_{L2}}{dt} = e_1$$

so our first state variable equation is

$$\frac{di_{L2}}{dt} = \frac{1}{L_2} e_1$$

If we sum currents into the node labeled n1 we get

$$i_a - i_{L2} - i_{C1} - i_{L1} = 0$$

This equation has our input (ia) and two state variable (iL2 and iL1) and the current through the capacitor. So from this we can get our second state equation

$$i_{C1} = C_1 \frac{de_1}{dt} = i_a - i_{L2} - i_{L1}$$

$$\frac{de_1}{dt} = \frac{1}{C_1}\left(i_a - i_{L2} - i_{L1}\right)$$

Our third, and final, state equation we get by writing an equation for the voltage across L_1 (which is e_2) in terms of our other state variables

$$e_2 = L_1 \frac{di_{L1}}{dt} = e_1 - R i_{L1}$$

$$\frac{di_{L1}}{dt} = \frac{1}{L_1}\left(e_1 - R i_{L1}\right)$$

We also need an output equation:

$$e_2 = e_1 - R i_{L1}$$

So our state space representation becomes

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} i_{L2} \\ e_1 \\ i_{L1} \end{bmatrix}$$

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u \qquad \mathbf{A} = \begin{bmatrix} 0 & \frac{1}{L_2} & 0 \\ -\frac{1}{C_1} & 0 & -\frac{1}{C_1} \\ 0 & \frac{1}{L_1} & -\frac{R}{L_1} \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{C_1} \\ 0 \end{bmatrix}$$

$$y = \mathbf{C}\mathbf{q} + Du \qquad \mathbf{C} = \begin{bmatrix} 0 & 1 & -R \end{bmatrix} \qquad D = 0$$

This technique does not always easily yield a set of state equations. In some cases it is easier to develop a transfer function model and convert this to a state space model. Transfer functions are discussed elsewhere.

Problems when developing a state space model from a system diagram

There are several cases when it is not so straightforward to develop a state space model from a system diagram. Some of these are discussed here.

Solution of State Space Problems

The state space representation of a system is a common and extremely powerful method of representing a system mathematically. This page only discusses how to develop the state space representation, the solution of state space problems are discussed elsewhere.

Transformations to other forms

Since state space is equivalent to the other representations, there must be a way to transform from one representation to another. These methods are discussed here.

References

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Transformation: Differential Equation ↔ State Space



Given a system differential equation it is possible to derive a state space model directly, but it is more convenient to go first derive the transfer function, and then go from the transfer function to the state space model.

Example: Diff Eq → State Space

Find a state space model for the system described by the differential equation:

$$\dddot{y} + a_1\ddot{y} + a_2\dot{y} + a_3 y = b_0\ddot{u} + b_1\dot{u} + b_2 u$$

Step 1: Find the transfer function using the methods described here (1DE ↔ TF)

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^2 + b_1 s + b_2}{s^3 + a_1 s^2 + a_2 s + a_3}$$

Step 2: Find a state space representation using the methods described here (TF ↔ SS). In this case we are using a CCF form).

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix}\mathbf{q} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \mathbf{C}\mathbf{q} + Du = \begin{bmatrix} b_2 & b_1 & b_0 \end{bmatrix}\mathbf{q} + 0 \cdot u$$

Example: State Space → Diff Eq

Repeat the previous example, but reverse the order (i.e., do step 2 (SS → TF) then step 1 (TF → SS)).

References

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Transformation: Transfer Function ↔ State Space

All transformation | Printable

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  - Other forms

Two of the most powerful (and common) ways to represent systems are the transfer function form and the state space form. This page describes how to transform a transfer function to a state space representation, and vice versa. Converting from state space form to a transfer function is straightforward because the transfer function form is unique. Converting from transfer function to state space is more involved, largely because there are many state space forms to describe a system.

State Space to Transfer Function

Consider the state space system:

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}u(t)$$
$$y(t) = \mathbf{C}\mathbf{q}(t) + Du(t)$$

Now, take the Laplace Transform (with zero initial conditions since we are finding a transfer function):

$$s\mathbf{Q}(s) = \mathbf{A}\mathbf{Q}(s) + \mathbf{B}U(s)$$
$$Y(s) = \mathbf{C}\mathbf{Q}(s) + DU(s)$$

We want to solve for the ratio of Y(s) to U(s), so we need so remove **Q**(s) from the output equation. We start by solving the state equation for **Q**(s)

$$s\mathbf{Q}(s) - A\mathbf{Q}(s) = \mathbf{B}U(s)$$
$$(s\mathbf{I} - \mathbf{A})\mathbf{Q}(s) = \mathbf{B}U(s)$$
$$\mathbf{Q}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}U(s) = \mathbf{\Phi}(s)\mathbf{B}U(s); \quad \text{where } \mathbf{\Phi}(s) = (s\mathbf{I} - \mathbf{A})^{-1}$$

The matrix **Φ**(s) is called the state transition matrix. Now we put this into the output equation

$$Y(s) = \mathbf{C}\mathbf{\Phi}(s)\mathbf{B}U(s) + DU(s)$$
$$= (\mathbf{C}\mathbf{\Phi}(s)\mathbf{B} + D)U(s)$$

Now we can solve for the transfer function:

$$H(s) = \frac{Y(s)}{U(s)} = \mathbf{C}\mathbf{\Phi}(s)\mathbf{B} + D = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D$$

Note that although there are many state space representations of a given system, all of those representations will result in the same transfer function (i.e., the transfer function of a system is unique; the state space representation is not).

Example: State Space to Transfer Function

Find the transfer function of the system with state space representation

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -4 & -2 \end{bmatrix}\mathbf{q} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}u$$
$$y = \mathbf{C}\mathbf{q} + Du = \begin{bmatrix} 5 & 1 & 0 \end{bmatrix} + 0 \cdot u$$

First find (s**I**-**A**) and the **Φ**=(s**I**-**A**)$^{-1}$ (note: this calculation is not obvious. Details are here). Rules for inverting a 3x3 matrix are here.

$$s\mathbf{I}-\mathbf{A}=\begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 3 & 4 & s+2 \end{bmatrix}$$

$$\mathbf{\Phi}=(s\mathbf{I}-\mathbf{A})^{-1}=\frac{\begin{bmatrix} s^2+2s+4 & 2+s & 1 \\ -3 & s(2+s) & s \\ -3s & -3-4s & s^2 \end{bmatrix}}{s^3+2s^2+4s+3}$$

Now we can find the transfer function

$$H(s)=\frac{Y(s)}{U(s)}=\mathbf{C\Phi B}+D$$
$$=\frac{s+5}{s^3+2s^2+4s+3}$$

To make this task easier, MatLab has a command (**ss2tf**) for converting from state space to transfer function.

```
>> % First define state space system
>> A=[0 1 0; 0 0 1; -3 -4 -2];
>> B=[0; 0; 1];
>> C=[5 1 0];
>> [n,d]=ss2tf(A,B,C,D)

n =
         0         0    1.0000    5.0000
d =
    1.0000    2.0000    4.0000    3.0000

>> mySys_tf=tf(n,d)
Transfer function:
       s + 5
----------------------
 s^3 + 2 s^2 + 4 s + 3
```

Example: State Space to Transfer Function (Symbolic)

Find the transfer function of the system with state space representation

$$\dot{\mathbf{q}}=\mathbf{Aq}+\mathbf{B}u=\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix}\mathbf{q}+\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}u$$
$$y=\mathbf{Cq}+Du=\begin{bmatrix} b_1 & b_0 & 0 \end{bmatrix}+0\cdot u$$

First find ($s\mathbf{I}$-$\mathbf{A}$) and the $\mathbf{\Phi}=(s\mathbf{I}-\mathbf{A})^{-1}$ (note: this calculation is not obvious. Details are here).

$$s\mathbf{I}-\mathbf{A}=\begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ a_3 & a_2 & s+a_1 \end{bmatrix}$$

$$\mathbf{\Phi}=(s\mathbf{I}-\mathbf{A})^{-1}=\frac{\begin{bmatrix} s^2+a_1s+a_2 & a_1+s & 1 \\ -a_3 & s(a_1+s) & s \\ -a_3s & -a_3-a_2s & s^2 \end{bmatrix}}{s^3+a_1s^2+a_2s+a_3}$$

Now we can find the transfer function

$$H(s) = \frac{Y(s)}{U(s)} = \mathbf{C\Phi B} + D$$
$$= \frac{b_0 s + b_1}{s^3 + a_1 s^2 + a_2 s + a_3}$$

Key Concept: Transforming from State Space to Transfer Function

Given a state space representation of a system

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}u(t)$$
$$y(t) = \mathbf{C}\mathbf{q}(t) + Du(t)$$

the transfer function is give by

$$H(s) = \frac{Y(s)}{U(s)} = \mathbf{C\Phi}(s)\mathbf{B} + D = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D$$

and the characteristic equation (i.e., the denominator of the transfer function) is

$$|s\mathbf{I} - \mathbf{A}|$$

Transfer Function to State Space

Recall that state space models of systems are not unique; a system has many state space representations. Therefore we will develop a few methods for creating state space models of systems.

Before we look at procedures for converting from a transfer function to a state space model of a system, let's first examine going from a differential equation to state space. We'll do this first with a simple system, then move to a more complex system that will demonstrate the usefulness of a standard technique.

First we start with an example demonstrating a simple way of converting from a single differential equation to state space, followed by a conversion from transfer function to state space.

Example: Differential Equation to State Space (simple)

Consider the differential equation with no derivatives on the right hand side. We'll use a third order equation, thought it generalizes to n^{th} order in the obvious way.

$$\dddot{y} + a_1\ddot{y} + a_2\dot{y} + a_3 y = b_0 u$$

For such systems (no derivatives of the input) we can choose as our n state variables the variable y and its first n-1 derivatives (in this case the first two derivatives)

$$q_1 = y$$
$$q_2 = \dot{y}$$
$$q_3 = \ddot{y}$$

Taking the derivatives we can develop our state space model

$$\dot{q}_1 = q_2 = \dot{y}$$
$$\dot{q}_2 = q_3 = \ddot{y}$$
$$\dot{q}_3 = \dddot{y} = -a_3 y - a_2\dot{y} - a_1\ddot{y} + b_0 u$$
$$= -a_3 q_1 - a_2 q_2 - a_1 q_3 + b_0 u$$
$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix}\mathbf{q} + \begin{bmatrix} 0 \\ 0 \\ b_0 \end{bmatrix}u$$
$$y = \mathbf{C}\mathbf{q} + Du = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\mathbf{q} + 0 \cdot u$$

*Note: For an nth order system the matrices generalize in the obvious way (**A** has ones above the main diagonal and the differential equation constants for the last row, **B** is all zeros with b_0 in the bottom row, **C** is zero except for the leftmost element which is one, and D is zero)*

Repeat Starting from Transfer Function

Consider the transfer function with a constant numerator (note: this is the same system as in the preceding example). We'll use a third order equation, thought it generalizes to n^{th} order in the obvious way.

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_0}{s^3 + a_1 s^2 + a_2 s + a_3}$$

$$\left(s^3 + a_1 s^2 + a_2 s + a_3\right) Y(s) = b_0 U(s)$$

For such systems (no derivatives of the input) we can choose as our n state variables the variable y and its first n-1 derivatives (in this case the first two derivatives)

$$q_1(t) = y(t) \qquad Q_1(s) = Y(s)$$
$$q_2(t) = \dot{y}(t) \qquad Q_2(s) = sY(s)$$
$$q_3(t) = \ddot{y}(t) \qquad Q_3(s) = s^2 Y(s)$$

Taking the derivatives we can develop our state space model (which is exactly the same as when we started from the differential equation)

$$sQ_1(s) = Q_2(s) = sY(s)$$
$$sQ_2(s) = Q_3(s) = s^2 Y(s)$$
$$sQ_3(s) = s^3 Y(s) = -a_1 s^2 Y(s) - a_2 sY(s) - a_3 Y(s) + b_0 u$$
$$= -a_1 s^2 Q_3(s) - a_2 sQ_2(s) - a_3 Q_1(s) + b_0 u$$

$$s\mathbf{Q}(s) = \mathbf{A}\mathbf{Q}(s) + \mathbf{B}U(s) == \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix} \mathbf{Q}(s) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} U(s)$$

$$Y(s) = \mathbf{C}\mathbf{Q}(s) + DU(s) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{Q}(s) + 0 \cdot U(s)$$

*Note: For an nth order system the matrices generalize in the obvious way (**A** has ones above the main diagonal and the coefficients of the denominator polynomial for the last row, **B** is all zeros with b_0 (the numerator coefficient) in the bottom row, **C** is zero except for the leftmost element which is one, and D is zero)*

If we try this method on a slightly more complicated system, we find that it initially fails (though we can succeed with a little cleverness).

Example: Differential Equation to State Space (harder)

Consider the differential equation with a single derivative on the right hand side.

$$\dddot{y} + a_1 \ddot{y} + a_2 \dot{y} + a_3 y = b_0 \dot{u} + b_1 u$$

We can try the same method as before:

$$q_1 = y$$
$$q_2 = \dot{y}$$
$$q_3 = \ddot{y}$$

$$\dot{q}_1 = q_2 = \dot{y}$$
$$\dot{q}_2 = q_3 = \ddot{y}$$
$$\dot{q}_3 = \dddot{y} = -a_3 y - a_2 \dot{y} - a_1 \ddot{y} + b_0 \dot{u} + b_1 u$$
$$= -a_3 q_1 - a_2 q_2 - a_1 q_3 + b_0 \dot{u} + b_1 u$$

The method has failed because there is a derivative of the input on the right hand, and that is not allowed in a state space model.

Fortunately we can solve our problem by revising our choice of state variables.

$$q_1 = y$$
$$q_2 = \dot{y}$$
$$q_3 = \ddot{y} - b_0 u$$

Now when we take the derivatives we get:

$$\dot{q}_1 = q_2 = \dot{y}$$
$$\dot{q}_2 = \ddot{y}$$
$$\dot{q}_3 = \dddot{y} - b_0 \dot{u} = -a_3 y - a_2 \dot{y} - a_1 \ddot{y} + b_1 u$$

The second and third equations are not correct, because $\ddot{y}$ is not one of the state variables. However we can make use of the fact:

$$q_3 = \ddot{y} - b_0 u, \quad \text{so}$$
$$\ddot{y} = q_3 + b_0 u$$

The second state variable equation then becomes

$$\dot{q}_2 = q_3 + b_0 u$$

In the third state variable equation we have successfully removed the derivative of the input from the right side of the third equation, and we can get rid of the $\ddot{y}$ term using the same substitution we used for the second state variable.

$$\dot{q}_3 = -a_3 y - a_2 \dot{y} - a_1 \ddot{y} + b_1 u$$
$$= -a_3 q_1 - a_2 q_2 - a_1 (q_3 + b_0 u) + b_1 u$$
$$= -a_3 q_1 - a_2 q_2 - a_1 q_3 + (b_1 - a_1 b_0) u$$

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix} \mathbf{q} + \begin{bmatrix} 0 \\ b_0 \\ b_1 - a_1 b_0 \end{bmatrix} u$$
$$y = \mathbf{C}\mathbf{q} + Du = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{q} + 0 \cdot u$$

The process described in the previous example can be generalized to systems with higher order input derivatives but unfortunately gets increasingly difficult as the order of the derivative increases. When the order of derivatives is equal on both sides, the process becomes much more difficult (and the variable "D" is no longer equal to zero). Clearly more straightforward techniques are necessary. Two are outlined below, one generates a state space method known as the "controllable canonical form" and the other generates the "observable canonical form (the meaning of these terms derives from Control Theory but are not important to us).

Controllable Canonical Form (CCF)

Probably the most straightforward method for converting from the transfer function of a system to a state space model is to generate a model in "controllable canonical form." This term comes from Control Theory but its exact meaning is not important to us. To see how this method of generating a state space model works, consider the third order differential transfer function:

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^2 + b_1 s + b_2}{s^3 + a_1 s^2 + a_2 s + a_3}$$

We start by multiplying by Z(s)/Z(s) and then solving for Y(s) and U(s) in terms of Z(s). We also convert back to a differential equation.

$$Y(s) = \left(b_0 s^2 + b_1 s + b_2\right) Z(s) \qquad y = b_0 \ddot{z} + b_1 \dot{z} + b_2 z$$
$$U(s) = \left(s^3 + a_1 s^2 + a_2 s + a_3\right) Z(s) \qquad u = \dddot{z} + a_1 \ddot{z} + a_2 \dot{z} + a_3 z$$

We can now choose z and its first two derivatives as our state variables

$$\begin{aligned} q_1 &= z & \dot{q}_1 &= \dot{z} = q_2 \\ q_2 &= \dot{z} & \dot{q}_2 &= \ddot{z} = q_3 \\ q_3 &= \ddot{z} & \dot{q}_3 &= \dddot{z} = u - a_1 \ddot{z} - a_2 \dot{z} - a_3 z \\ & & &= u - a_1 q_3 - a_2 q_2 - a_3 q_1 \end{aligned}$$

Now we just need to form the output

$$\begin{aligned} y &= b_0 \ddot{z} + b_1 \dot{z} + b_2 z \\ &= b_0 q_3 + b_1 q_2 + b_2 q_1 \end{aligned}$$

From these results we can easily form the state space model:

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix} \mathbf{q} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$
$$y = \mathbf{C}\mathbf{q} + Du = \begin{bmatrix} b_2 & b_1 & b_0 \end{bmatrix} \mathbf{q} + 0 \cdot u$$

In this case, the order of the numerator of the transfer function was less than that of the denominator. If they are equal, the process is somewhat more complex. A result that works in all cases is given below; the details are here. For a general n^{th} order transfer function:

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

the controllable canonical state space model form is

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u; \quad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix}; \qquad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$y = \mathbf{C}\mathbf{q} + Du \quad \mathbf{C} = \begin{bmatrix} b_n - a_n b_0 & b_{n-1} - a_{n-1} b_0 & \cdots & b_2 - a_2 b_0 & b_1 - a_1 b_0 \end{bmatrix} \quad D = b_0$$

Key Concept: Transfer function to State Space (CCF)

For a general n^{th} order transfer function:

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

the controllable canonical state space model form is

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u; \quad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$y = \mathbf{C}\mathbf{q} + Du \quad \mathbf{C} = \begin{bmatrix} b_n - a_n b_0 & b_{n-1} - a_{n-1} b_0 & \cdots & b_2 - a_2 b_0 & b_1 - a_1 b_0 \end{bmatrix} \quad D = b_0$$

Observable Canonical Form (OCF)

Another commonly used state variable form is the "observable canonical form." This term comes from Control Theory but its exact meaning is not important to us. To understand how this method works consider a third order system with transfer function:

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^2 + b_1 s + b_2}{s^3 + a_1 s^2 + a_2 s + a_3}$$

We can convert this to a differential equation and solve for the highest order derivative of y:

$$\begin{aligned} \left(s^3 + a_1 s^2 + a_2 s + a_3\right) Y(s) &= \left(b_0 s^2 + b_1 s + b_2\right) U(s) \\ s^3 Y(s) &= \left(b_0 s^2 + b_1 s + b_2\right) U(s) - \left(a_1 s^2 + a_2 s + a_3\right) Y(s) \\ \dddot{y} &= b_0 \ddot{u} + b_1 \dot{u} + b_2 u - a_1 \ddot{y} - a_2 \dot{y} - a_3 y \end{aligned}$$

Now we integrate twice (the reason for this will be apparent soon), and collect terms according to order of the integral:

$$\begin{aligned} \dot{y} &= b_0 u + b_1 \int u \cdot dt + b_2 \int\int u \cdot dt \cdot dt - a_1 y - a_2 \int y \cdot dt - a_3 \int\int y \cdot dt \cdot dt \\ &= b_0 u - a_1 y + \int \left(b_1 u - a_2 y\right) dt + \int\int \left(b_2 u - a_3 y\right) \cdot dt \cdot dt \end{aligned}$$

Choose the output as our first state variable

$$q_1 = y \qquad \dot{q}_1 = \dot{y} = b_0 u - a_1 y + \int \left(b_1 u - a_2 y\right) dt + \int\int \left(b_2 u - a_3 y\right) \cdot dt \cdot dt$$

Looking at the right hand side of the differential equation we note that $y=q_1$ and we call the two integral terms q_2:

$$q_2 = \int \left(b_1 u - a_2 y\right) dt + \int\int \left(b_2 u - a_3 y\right) dt \cdot dt$$

so

$$\dot{q}_1 = \dot{y} = b_0 u - a_1 q_1 + q_2$$

This is our first state variable equation.

Now let's examine q_2 and its derivative:

$$\begin{aligned} q_2 &= \int \left(b_1 u - a_2 y\right) dt + \int\int \left(b_2 u - a_3 y\right) dt \cdot dt \\ \dot{q}_2 &= b_1 u - a_2 y + \int \left(b_2 u - a_3 y\right) dt \end{aligned}$$

Again we note that $y=q_1$ and we call the integral terms q_3:

$$q_3 = \int \left(b_2 u - a_3 y\right) dt$$

so

$$\dot{q}_2 = b_1 u - a_2 q_1 + q_3$$

This is our second state variable equation.

Now let's examine q_3 and its derivative:

$$q_3 = \int (b_2 u - a_3 y)\, dt$$
$$\dot{q}_3 = b_2 u - a_3 y$$
$$= b_2 u - a_3 q_1$$

This is our third, and last, state variable equation.

Our state space model now becomes:

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u = \begin{bmatrix} -a_1 & 1 & 0 \\ -a_2 & 0 & 1 \\ -a_3 & 0 & 0 \end{bmatrix} \mathbf{q} + \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix} u$$
$$y = \mathbf{C}\mathbf{q} + Du = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{q} + 0 \cdot u$$

In this case, the order of the numerator of the transfer function was less than that of the denominator. If they are equal, the process is somewhat more complex. A result that works in all cases is given below; the details are here. For a general n^{th} order transfer function:

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

the observable canonical state space model form is

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u; \quad \mathbf{A} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & 0 & \vdots \\ \vdots & \vdots & 0 & \ddots & 0 \\ -a_{n-1} & 0 & \vdots & \cdots & 1 \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ \vdots \\ b_{n-1} - a_{n-1} b_0 \\ b_n - a_n b_0 \end{bmatrix}$$

$$y = \mathbf{C}\mathbf{q} + Du \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \end{bmatrix} \qquad D = b_0$$

Key Concept: Transfer function to State Space (OCF)

For a general n^{th} order transfer function:

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

the observable canonical state space model form is

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u; \quad \mathbf{A} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & 0 & \vdots \\ \vdots & \vdots & 0 & \ddots & 0 \\ -a_{n-1} & 0 & \vdots & \cdots & 1 \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ \vdots \\ b_{n-1} - a_{n-1} b_0 \\ b_n - a_n b_0 \end{bmatrix}$$

$$y = \mathbf{C}\mathbf{q} + Du \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \end{bmatrix} \qquad D = b_0$$

Other forms

There are many other forms that are possible. For example MATLAB uses a variant of the controllability canonical form in its "`ss2tf`" function.

References

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Comments? Questions? Suggestions? Corrections?
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Transformation: State Space ↔ State Space

All transformation Printable

A state variable representation of a system is not unique. In fact there are infinitely many representations. Methods for transforming from one set of state variables to another are discussed below, followed by an example.

Consider the state space representation

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t)$$
$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{u}(t)$$

We can define a new set of independent variables (i.e., **T** is invertible)

$$\hat{\mathbf{q}} = \mathbf{T}\mathbf{q}$$

Though it may not be obvious we can use this new set of variables as state variables. Start by solving for q

$$\mathbf{q} = \mathbf{T}^{-1}\hat{\mathbf{q}}$$

(note: some textbooks use the matrix $\mathbf{P}=\mathbf{T}^{-1}$ *to define the transformation)*

We can now rewrite the state space model by replacing q in the original equations

$$\mathbf{T}^{-1}\dot{\hat{\mathbf{q}}} = \mathbf{A}\mathbf{T}^{-1}\hat{\mathbf{q}} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{T}^{-1}\hat{\mathbf{q}} + \mathbf{D}\mathbf{u}$$

Multiply the top equation by **T** to solve for $\dot{\hat{\mathbf{q}}}$

$$\dot{\hat{\mathbf{q}}} = \mathbf{T}\mathbf{A}\mathbf{T}^{-1}\hat{\mathbf{q}} + \mathbf{T}\mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{T}^{-1}\hat{\mathbf{q}} + \mathbf{D}\mathbf{u}$$

We recognize this as a state space representation

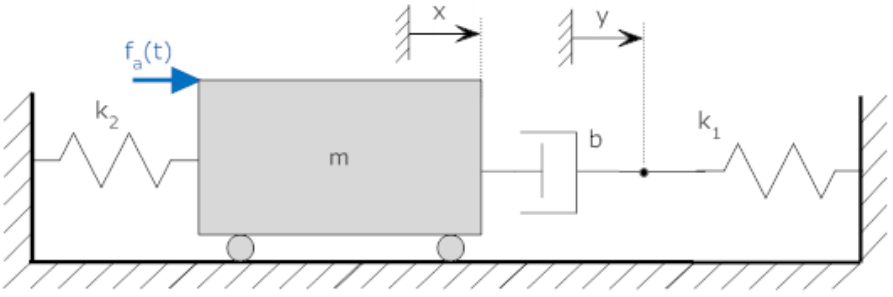
$$\dot{\hat{\mathbf{q}}} = \hat{\mathbf{A}}\hat{\mathbf{q}} + \hat{\mathbf{B}}\mathbf{u}$$
$$\mathbf{y} = \hat{\mathbf{C}}\hat{\mathbf{q}} + \hat{\mathbf{D}}\mathbf{u}$$

with

$$\hat{\mathbf{A}} = \mathbf{T}\mathbf{A}\mathbf{T}^{-1}$$
$$\hat{\mathbf{B}} = \mathbf{T}\mathbf{B}$$
$$\hat{\mathbf{C}} = \mathbf{C}\mathbf{T}^{-1}$$
$$\hat{\mathbf{D}} = \mathbf{D}$$

Example: State Space to State Space

Consider the system below:



We have shown that the choice of state variables

$$\mathbf{q} = \begin{bmatrix} x \\ \dot{x} \\ y \end{bmatrix}$$

yields the state space system

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{B}u \qquad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{k_1+k_2}{m} & 0 & \frac{k_1}{m} \\ \frac{k_1}{b} & 0 & -\frac{k_1}{b} \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \end{bmatrix}$$

$$y = \mathbf{C}\mathbf{q} + Du \qquad \mathbf{C} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \qquad D = 0$$

If we want a set of state variables that includes the position of the center of the dashpot ½(x+y) as well as the extension of the dashpot (y-x), find a new state variable representation.

Solution:

Our new state variables are:

$$\hat{q} = \begin{bmatrix} \frac{1}{2}(x+y) \\ y-x \\ \dot{x} \end{bmatrix}$$

which we can write in terms of the old variables as:

$$\hat{q} = \begin{bmatrix} \frac{1}{2}(x+y) \\ y-x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ y \end{bmatrix} = \mathbf{T}\mathbf{q} \qquad \mathbf{T} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

It can be shown that

$$\mathbf{T}^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \\ 1 & \frac{1}{2} & 0 \end{bmatrix}$$

so

$$\dot{\hat{\mathbf{q}}} = \hat{\mathbf{A}}\hat{\mathbf{q}} + \hat{\mathbf{B}}u$$
$$\mathbf{y} = \hat{\mathbf{C}}\hat{\mathbf{q}} + \hat{\mathbf{D}}u$$

with

$$\hat{\mathbf{A}} = \mathbf{T}\mathbf{A}\mathbf{T}^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ \frac{-k_2}{m} & \frac{-b}{m} & \frac{b}{m} \\ 0 & 1 & -\frac{k_1}{b} \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \\ 1 & \frac{1}{2} & 0 \end{bmatrix} = \begin{bmatrix} -\frac{k_1}{2b} & -\frac{k_1}{4b} & 1 \\ -\frac{k_1}{b} & -\frac{k_1}{2b} & 0 \\ \frac{b-k_2}{m} & \frac{b+k_2}{m} & \frac{b}{m} \end{bmatrix}$$

$$\hat{\mathbf{B}} = \mathbf{T}\mathbf{B} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \frac{1}{m} \end{bmatrix} = \begin{bmatrix} \frac{1}{2m} \\ \frac{1}{m} \\ 0 \end{bmatrix}$$

$$\hat{\mathbf{C}} = \mathbf{C}\mathbf{T}^{-1} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \\ 1 & \frac{1}{2} & 0 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \end{bmatrix}$$

$$\hat{\mathbf{D}} = \mathbf{D} = 0$$

Matlab Solution



Matlab code, html

Use MatLab to convert from state space to state space (symbolic)

This script requires the MatLab's symbolic toolbox

```
% Start by clearing everything
clear all
clc
```

Declare all symbolic variables

```
syms k1 k2 b m
```

Define state space system

References

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Comments? Questions? Suggestions? Corrections?
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